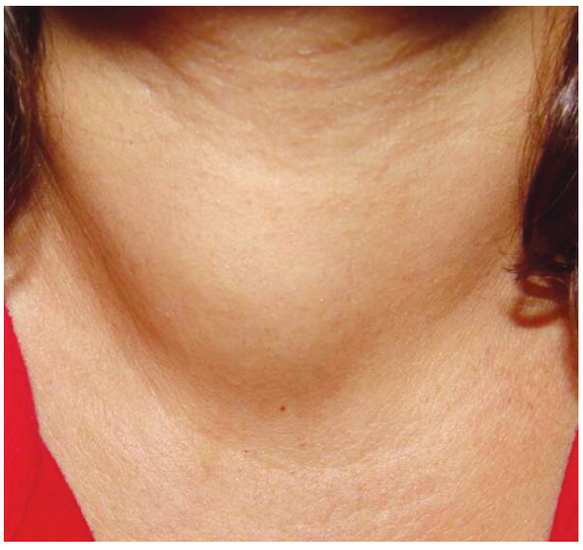


Figure 9.16 goiter

9.3.4. Fluorine

Fluorine exists in the body as compounds called fluorides.It is used as an anti-coagulant during collection of blood for the determination of blood sugar.

Functions

1. Fluorine is required in traces for the development of bones and teeth.
2. It is necessary for the prevention of dental caries.
3. It is used in combination with vitamin D for the treatment of osteoporosis.

Sources

The body receives fluorine mainly from drinking water, in which concentration varies with soil content. Some sea fish and tea also contain small amounts of fluorine.

Absorption and excretion:

Absorption of fluoride is via the small intestine into the blood stream. Most of the fluorides that are not retained by the bones and teeth are excreted rapidly into the urine. It is also excreted through sweat and intestine.

Deficiency

The absence of fluorine in the diet causes dental caries.

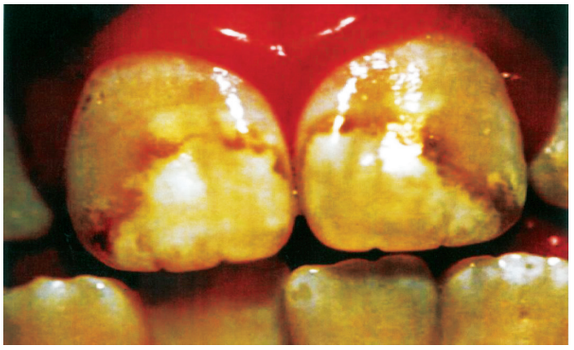


Figure 9.17 Dental Fluorosis

9.3.5. Zinc

Adult human body contains approximately 1 to 2 grams of zinc. Like iron, zinc is absorbed according to the body needs. Prostate gland is very rich in zinc.

Functions

1. Zinc forms an integral part of several metallo- enzymes in the body.
2. Zinc is essential for growth and reproduction.
3. It has role in maintaining the plasma concentration of vitamin A.
4. Zinc is essential for the storage of insulin in the β-islet cells.
5. Zinc is necessary for wound healing.

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